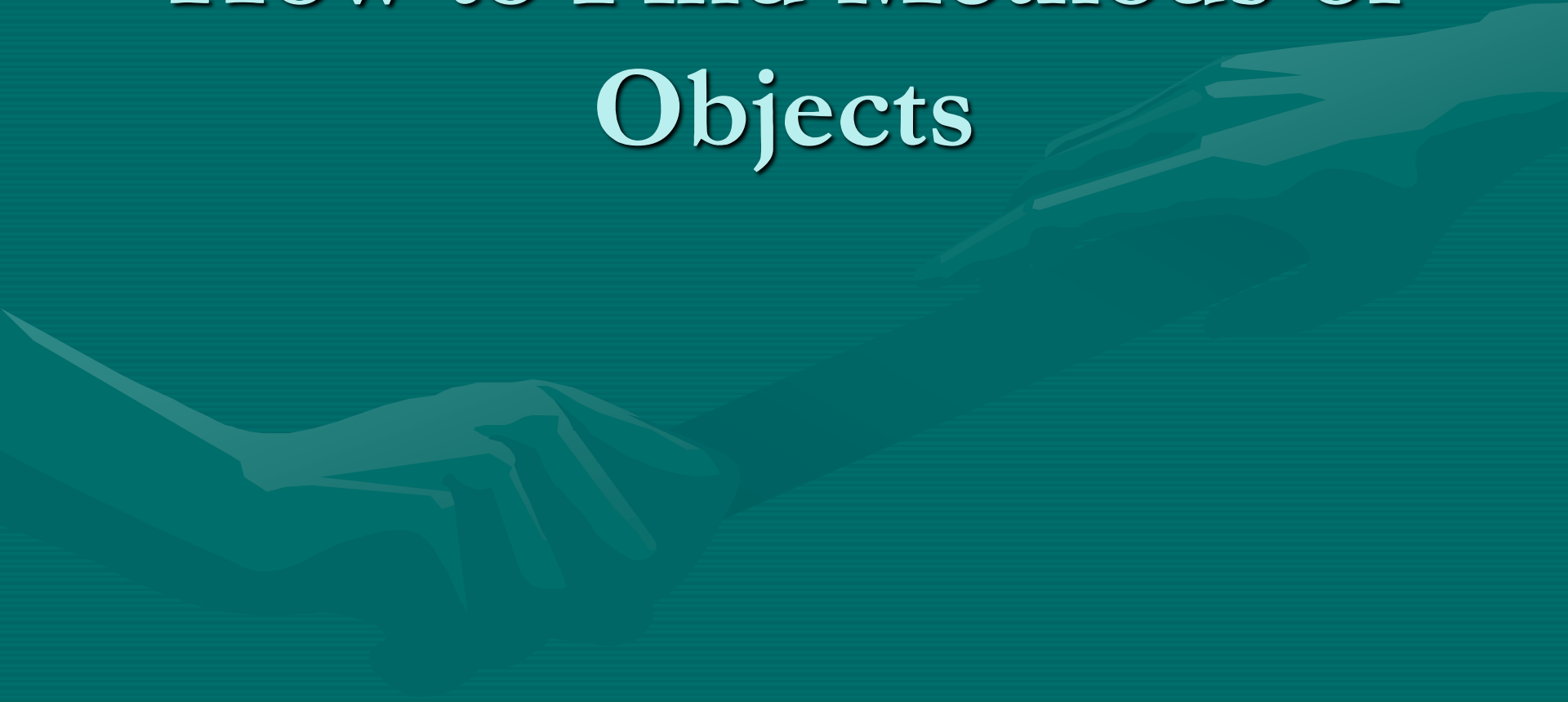


How to Find Methods of Objects



How to find methods of object

- Two approaches
 - **maximize**
 - when you design a class library which is intended to be used for generic purpose
 - You do not know what is a complete set of methods
 - You tend to maximize them. Do as much as you can. Basically you guess how other applications will use your class
 - If your task is to complete an application, coding methods in this way will never work. You will find you produce lots of behaviors which are never called by others. This is sometimes called the “C++ newbie syndrome”

- **Minimize it**

- only implement the needed methods
- each method is carefully planned
- each method is called by what classes is pre-determined.
- each written method is definitely invoked
- suitable for a project which has deadline with precise goals.

The problem of OO programs

- OOAD requires the behaviors to be placed under the right class to reduce maintenance problems
- **Side effect** are:
 - “sometimes the execution flow (logics) are obscured by inheritance and polymorphism
 - A function feature may be completed by several classes via several methods. It is not intuitive as traditional C programs.
(that is why UML introduces sequence diagram to visualize the execution flow of a scenario or test case)

UML sequence diagram

- Let's take a look at sequence diagram



Finding the object behaviors

- There are many methodology exist in OOAD community
 - CRC cards
 - **test case based**
 - CRUD methods



Use Case Based Approach

- Use Cases
 - The de facto standard UML way to specify software
 - describes what feature (behaviors) a program should complete.
 - Use case is the entry point of execution flows.
 - each use case should be completed by object methods (via collaboration)

Use Case Based Approach

- Precondition
 - A close to 90% correct class diagram (without methods yet) should be ready.
- Algorithm
 - for each test case
 1. write down pseudo code
 2. for each part of pseudo code
 1. determine which class should take over or related
 2. create methods for the class.
 3. evaluate the code maintainability
 3. until each part of the pseudo code is placed

The basics of the case study

- The event flow of a general window system
- please write down the pseudo code for some test cases
- distribute pseudo code into class methods.
- let's exercise

Exercise

- Your task is to draw sequence diagram to two test cases in your case study.
- In order to draw the sequence diagram, you need to find and create the methods first.